

1a. Develop a program to read the student details like USN, name and marks of three subjects .Display the student details, total marks and percentage with suitable message

Program:

```
USN=input(" Enter Usn ")
name=input(" Enter Name:")
print(" Enter Marks Of 3 Subjects ")
marks=[] # define an Empty list marks
for i in range(3):
    m=int(input(" Enter The Marks :"))
    marks.append((m)) # load the values to list marks using append method
print(marks)
total=sum(marks) # calculate the total by using sum() function and store it in variable total
per=int(100*total)/300 # calculate the percentage and store it in variable per
print(USN,name,total,per)
```

Output:

```
Enter Usn 4BD22IS005
Enter Name: hafsa
Enter Marks Of 3 Subjects
Enter The Marks :90
[90]
Enter The Marks :60
[90, 60]
Enter The Marks :80
[90, 60, 80]
4BD22IS005 hafsa 230 76.66666666666667
```


1b.Develop a program to read the name and year of birth of a person .Display whether the person is senior citizen or not.

Program:

```
import datetime
name=input('Enter your name')
birthyear=int(input('Enter the year'))
today= datetime.date.today() #used inbuilt function to read today's date (format yyyy-mm-dd)
print(today)
currentyear=today.strftime('%Y') # strftime (string format time)
print(currentyear)
age=int(currentyear)-int(birthyear)
print('Your age is' + age)
if age>=60:
print('you are a senior citizen')
else:
print('you are not a senior citizen')
```

Output:

- Enter your name punith
Enter the year 2004
2023-01-11
2023
Your age is 19
You are not a senior citizen
- Enter your name sreenivas
Enter the year 1945
2023-01-11
2023
Your age is 78
You are a senior citizen

2a. Develop a program to generate Fibonacci sequence of length (N).Read N from the console.

```
nterms = int(input("How many terms? "))
# first two terms
n1, n2 = 0, 1
count = 0
# check if the number of terms is valid
if nterms <= 0:
    print("Please enter a positive integer")
# if there is only one term, return n1
elif nterms == 1:
    print("Fibonacci sequence upto",nterms,":")
    print(n1)
# generate fibonacci sequence
else:
    print("Fibonacci sequence:")
    while count < nterms:
        print(n1)
        nth = n1 + n2
        # update values
        n1 = n2
        n2 = nth
        count += 1
```

Output:

How many terms? 7

Fibonacci sequence:

0

1

1

2

3

5

8

2b. Write a function to calculate a factorial of a number. Develop a program to compute binomial coefficient (Given N and R)

```
def fact(n): # define a function Factorial(Fact)
    if n==0:
        return 1
    else:
        return n*fact(n-1) # Recursion (recursive function can be defined that calls itself)

n=int(input('Enter n'))
r=int(input('Enter r'))
if n<0 or r<0:
    print('invalid')
else:
    fn=fact(n) #Function call to calculate n!
    Print('Factorial of n is',fn)
    fr=fact(r) #Function call to calculate r!
    Print('Factorial of r is',fr)
    fnr=fact(n-r) #Function call to calculate (n-r)!
    Print('Factorial of n-r is',fnr)
    BC=fn/(fr*fnr) # computing binomial coefficient nCr= n!/r!*(n-r)!
    print ('binomial coefficient is',BC)
```

Output:

Enter n 6

Enter r 3

Factorial of n is 720

Factorial of r is 6

Factorial of n-r is 6

Binomial coefficient is 20.0

3. Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable messages

```
import math
def mean(data):                                # Function to find the mean
    n = len(data)
    mean = sum(data) / n
    return mean

def variance(data):                            # Function to find the variance
    n = len(data)
    mean = sum(data) / n
    deviations = [(x - mean) ** 2 for x in data]
    variance = sum(deviations) / n
    return variance

def stdev(data):                              # Function to find the Standard Deviation
    var = variance(data)
    std_dev = math.sqrt(var)
    return std_dev

listA = []                                    # creating an empty list
n = int(input("Enter number of elements : ")) # number of elements as input

for i in range(0, n):                        # iterating till the range
    element = int(input())
    listA.append(element) # adding the element

print(listA) # Display the elements in the list
data=listA
print("Mean of the sample is % s " % (mean(data)))
print("variance of the sample is % s " % (variance(data)))
print("Standard Deviation of the sample is % s "% (stdev(data)))
```

Output:

```
Enter number of elements
10
4,8,6,5,3,2,8,9,2,5
[4,8,6,5,3,2,8,9,2,5]
Mean of the sample is 5.2
Variance of the sample is 5.76
Standard deviation of sample is 2.4
```


4. Read a multi-digit number (as chars) from the console. Develop a program to print frequency of each digit with suitable example.

Program:

```
frequency={}
Print("Enter the multi digit number")
digit=input()
For i in digit:
    if i not in frequency :
        frequency[i]=1
    else:
        frequency[i]=frequency[i]+1
print (frequency)
```

Output:

```
Enter the multi digit number
12345
{'1': 1, '2': 1, '3': 1, '4': 1, '5': 1}
Enter the multi digit number
12221333344
{'1': 2, '2': 3, '3': 4, '4': 2}
```


5. Develop a program to print 10 most frequently appearing words in a text file [Hint: use dictionary with distinct words and their frequency of occurrences. Sort the dictionary in the reverse order of frequency and display dictionary slice of first 10 items]

Program:

```
file=open("D://Words.txt")          # Provide the path of note pad file
text=file.read()                    # Read the file and store it in text
#print(text)
words=text.split()                  # split the text considering space between words
frequency={} # Create a dictionary frequency
for word in words:
    if word not in frequency:        # this part of the code will check whether the word is already
        frequency[word]=1           # present in the dictionary if present it will execute else
    else:                            # condition and increment frequency , if not present it will
        frequency[word]+=1          # add word to the dictionary
#print(frequency)
most_frequent = dict(sorted(frequency.items(),key=lambda elem: elem[1],reverse=True)) # sort in descending order
#print(most_frequent)
out = dict(list(most_frequent.items())[0: 10]) # select the top 10 most frequently used words
print("1st 10 Most Frequent words are:"+str(out)) # print the output
```

Prerequisites: Create a Text file (Words.txt) and add the Content

File name: Words.txt

Once upon a time there lived a poor widow and her son Jack. One day, Jack's mother told him to sell their only cow. Jack went to the market and on the way he met a man who wanted to buy his cow

Output:

1st 10 Most Frequent words are :{'a': 3, 'to': 3, 'and': 2, 'Jack': 2, 'the': 2, 'Once': 1, 'upon': 1, 'time': 1, 'there': 1, 'lived': 1}